

The diagram illustrates the Tucker decomposition of a 3D tensor T . On the left, a purple cube represents the tensor T , with the label T inside and the dimensionality $\in \mathbb{R}^{n \times m \times p}$ below it. To the right of the cube is an equals sign, followed by a sum of r terms. Each term consists of a blue vertical line segment labeled a_i (for $i=1, 2, \dots, r$), an orange horizontal line segment labeled b_i , and a green diagonal line segment labeled c_i . The segments are connected at their endpoints to form a small L-shaped structure. The terms are separated by plus signs, with an ellipsis indicating the continuation of the sum.

$$T \in \mathbb{R}^{n \times m \times p} = \begin{matrix} \text{green } c_1 \\ \text{orange } b_1 \\ \text{blue } a_1 \end{matrix} + \begin{matrix} \text{green } c_2 \\ \text{orange } b_2 \\ \text{blue } a_2 \end{matrix} + \dots + \begin{matrix} \text{green } c_r \\ \text{orange } b_r \\ \text{blue } a_r \end{matrix}$$